

if/else

Executes a block of code if a condition is true; otherwise, executes another block.

Syntax Example:

```
$age = 20;

if $age >= 18 {
    echo "Adult";
} Else {
    echo "Minor";
}
```

Result: Adult

elseif

Adds an additional condition to an if/else chain.

Syntax Example:

```
$score = 75;

if $score >= 90 {
    echo "Excellent";
} elseif $score >= 50 {
    echo "Pass";
} else {
    echo "Fail";
}
```

Result: Pass

Switch

Selects one block of code to run from multiple options based on a variable's value.

Syntax Example:

```
$day = "Tuesday";

switch $day {
    case "Monday":
        echo "Start of the week";
        Break;
    case "Tuesday":
        echo "Second day";
        Break;
    default:
        echo "Another day";
}
```

Result: Second day

Match

Returns a value based on a condition; similar to switch but strict comparison and returns a value.

Syntax Example:

```
$day = "Tuesday";

$result = Match($day) {
    "Monday" => "Start of the week",
    "Tuesday" => "Second day",
    default => "Another day"
};

echo $result;
```

Result: Second day

Ternary Operator

Shorthand for simple if/else statements.

Syntax Example:

```
$age = 20;
```

```
“(Condition) ? If true : if false”  
echo $age >= 18 ? "Adult" : "Minor";
```

Result: Adult

Null Coalescing Operator

Returns the first value that exists and is not null.

Syntax Example:

```
$username = $_GET['user'] ?? "Guest";  
  
echo $username;
```

Result: Guest *(if \$_GET['user'] is not set)*

Null Safe Operator

Safely accesses properties or methods on an object that might be null.

Syntax Example:

```
$user = null;  
  
echo $user?->name;
```

Result: *(no error, outputs nothing)*

Logical Operators

Used to combine or negate conditions.

Syntax Example:

```
$age = 20;  
$has_ID = true;  
  
If $age >= 18 && $hasID {  
    echo "Allowed";  
} else {  
    echo "Not allowed";  
}
```

Result: Allowed

Notes :

- && → AND
- || → OR
- ! → NOT

Comparison Operators

Compare values to determine true/false.

Syntax Example:

```
$a = 5;  
$b = 10;  
  
If $a < $b {  
    echo "Yes";  
} else {  
    echo "No";  
}
```

Result: Yes

Notes :

- == → equals
- === → identical (value + type)

- $\neq \rightarrow$ not equal
- $\neq \rightarrow$ not identical
- $<, >, \leq, \geq \rightarrow$ comparison